

Printing a W-Shaped Pattern

Mastering Pattern Problems in Python

Question 2: 2025 Jan OPPE 1 - Set 1 – Section 3

Understanding the Problem

The Goal

Given an integer $n > 0$, print a W-shaped pattern with vertical bars on the edges and diagonal slashes that move inward.

Input/Output

This is a pure I/O problem. You read an integer from input and print the pattern directly to output without returning values.

Key Characters

The pattern uses vertical bars `|`, forward slashes `/`, backslashes `\`, and spaces for positioning.

Pattern Examples

Example 1

Input: 1

```
|/\|
```

Example 2

Input: 2

```
| /\|  
|/ \|
```

Example 3

Input: 3

```
| /\|  
|/ \|  
|/ \|
```

Notice how the slashes move diagonally inward as the pattern grows.
Each row has a specific spacing pattern that determines the W shape.

Example 3: Row Breakdown

Let's examine the pattern for $n = 3$, showing how each row builds the W shape:

1

Row 0 (Top)

Outermost spacing - largest gaps on the sides, smallest middle gap

| /\ |

2

Row 1 (Middle)

Medium spacing - sides get closer, middle gap widens

| / \ |

3

Row 2 (Bottom)

Closest spacing - sides almost touch, largest middle gap

|/ \|

Visualizing the Pattern Logic

The key insight is recognizing that the slashes move diagonally inward from top to bottom. Each row has a predictable spacing pattern that depends on its position index.

Analyzing a Single Row

Understanding one row's structure unlocks the entire solution. For any row index `i`:

01

Calculate Left Spaces

Number of spaces before the forward slash

02

Calculate Middle Spaces

Number of spaces between the forward and backslash

03

Build the Pattern

Combine all elements with vertical bars at the edges

The Complete Formula

Row Structure

For row index i :

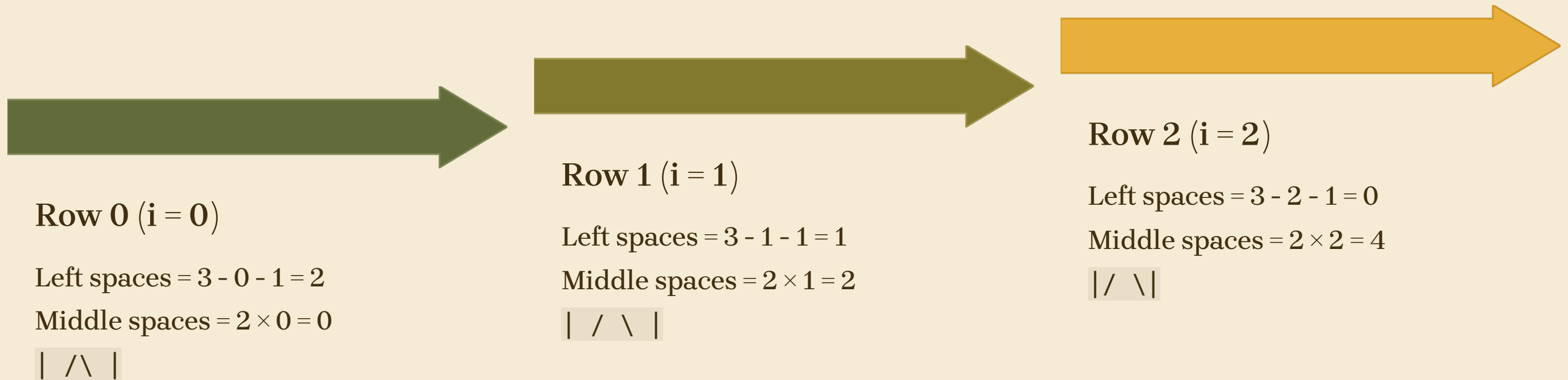
```
| + left_spaces + / +  
middle_spaces + \ +  
left_spaces + |
```

Spacing Calculations

- **Left spaces** = $n - i - 1$
- **Middle spaces** = $2 \times i$

The left spaces decrease as we move down rows, while middle spaces increase—creating the diagonal inward movement.

Deriving the Pattern: $n = 3$



The formula matches perfectly! Notice how left spaces decrease while middle spaces increase, creating the W shape.

Step-by-Step Solution Plan



Read Input

Get integer n from user input



Loop Rows

Iterate from $i = 0$ to $n-1$



Calculate Spacing

Compute left and middle spaces using formulas



Print Row

Format and output each row

 **Key Advantage:** This approach uses a single loop with direct calculations—no nested loops needed!

Python Implementation

```
n = int(input())
for i in range(n):
    left_spaces = n - i - 1
    middle_spaces = 2 * i
    print(
        "|" +
        " " * left_spaces +
        "/" +
        " " * middle_spaces +
        r"\" +
        " " * left_spaces +
        "|"
    )
```

Common Mistakes to Avoid

- Forgetting to escape backslash: use `\\`
- Wrong middle spacing formula
- Using nested loops unnecessarily
- Adding extra spaces in wrong positions

Why Students Struggle

Students fail when they start coding without visualizing, guess spacing patterns, or don't derive the formula first.

The Logic: For each row i , $\text{left} = n - i - 1$ and $\text{middle} = 2 \times i$.